

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, 2D image slices are acquired in one pre-defined direction one after the other, and the final 3D volume is created by stacking the 2D slices. The voxel sizes are given as in-plane (two values, the pixel size within the imaging plane) and through-plane (one value, the slice thickness). Recall that the field-of-view is the real-world size of the entire image, i.e. image size $\times$ voxel size. Based on this please compute the following.
 - (a) A *coronal* acquisition with voxel size in-plane $0.9 \times 0.9 \text{ mm}^2$ with 260 pixels in both directions and voxel size through-plane 3.0 mm with 64 pixels is acquired. What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?
 - (b) The protocol has to be changed so that the anterior-to-posterior coverage is at least 210 mm, while the slice thickness stays at 3.0 mm. How many slices have to be acquired?
 - (c) You want to resample the volume of part (a) to isotropic voxels of $1.0 \times 1.0 \times 1.0 \text{ mm}^3$. What are the resize factors and the resulting image size? In which of the three directions is the resampling least justified by the information actually contained in the data, and why?
2. Histogram equalization improves the contrast of an image by finding a monotonically increasing intensity mapping $i \mapsto f(i)$ that flattens the histogram. Let $p(i)$ denote the probability density

function of the intensities, $P(i) = \int_0^i p(j) dj$ the cumulative distribution function, and i_{max} the maximum intensity. Which of the two mappings below is the one that flattens the histogram? Please circle the correct choice.

$$f(i) = i_{max} p(i) \qquad f(i) = i_{max} P(i) = i_{max} \int_0^i p(j) dj$$

3. When a Markov random field prior is used, segmentation can be written either as a posterior maximization or as an energy minimization. I is the image, c the labelling, $G(x)$ the neighbours of pixel x , and Z a normalization constant.

$$\max_c \log \left[\prod_{x \in \Omega} p(I(x) | c(x)) \right] + \log \left[\frac{1}{Z} \exp\{-E(c)\} \right]$$

$$\min_c \underbrace{\sum_{x \in \Omega} g(I(x) | \theta_{c(x)})}_{\text{unary term}} + \underbrace{\sum_{x \in \Omega} \sum_{y \in G(x)} d(c(x), c(y))}_{\text{pairwise term}}$$

- (a) Please indicate which term corresponds to which term in the two formulations.
- (b) Which theorem guarantees that a prior satisfying the Markov property can be written as a Gibbs distribution $\frac{1}{Z} \exp\{-E(c)\}$ with a locally defined energy E ?
4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.
- (a) True False A rigid-body transformation in 3D has 6 degrees of freedom: three rotation angles and three translations.
 - (b) True False For *isotropic* scaling, $SRx = RSx$, i.e. the order in which rotation and scaling are applied does not matter.
 - (c) True False In homogeneous coordinates, a 3D affine transformation is written as a 3×3 matrix.
 - (d) True False The determinant of the Jacobian of a transformation is larger than 1 in the regions that the transformation compresses.

5. A free-form deformation (FFD) model with cubic B-splines is used for the non-linear registration of two 3D images. The control points are placed on a uniform grid of $10 \times 12 \times 8$ points. How many parameters does this transformation model have, and how does this compare with the number of parameters of a 3D affine transformation?
6. The N3 algorithm corrects the bias field in MRI. It works in the log domain, where the observed image is $j(x) = b(x) + i(x)$ with b the bias field and i the bias-free image, and where the bias field is assumed to be independent of the image intensities. Each iteration alternates between two estimation problems. Please briefly describe what each of them does, and explain why the algorithm has to iterate with narrow Gaussians instead of correcting the bias in a single step. You do not need to write down the equations.
7. Which of the following statements on the loss metrics used for intensity-based registration is **inaccurate**?
- ☐ (A) The sum of squared differences assumes that the intensities of the two images are directly comparable, which restricts it to intra-modality registration.
 - ☐ (B) Normalized cross-correlation attains its maximum magnitude when the two images are related by a linear intensity relationship that is the same at every pixel; both positive and negative correlations indicate alignment, which is why the squared value is used in the loss.
 - ☐ (C) Mutual information is maximized when the joint intensity distribution equals the product of the two marginals, that is, when the intensities of the two images are statistically independent.
 - ☐ (D) Mutual information can capture more complicated, non-linear relationships between the intensities than normalized cross-correlation, but it is harder to compute.

8. Which of the following statements about interpretability and explainable AI (XAI) in medical image analysis is **incorrect**?
- ☐ (A) Black-box, or model-agnostic, methods such as LIME require no access to the internals of the model, whereas gradient-based methods such as Grad-CAM are white-box methods.
 - ☐ (B) Grad-CAM generalizes CAM: the importance weight of a feature map is the global average of the gradients of the class score with respect to that feature map, which makes it applicable to any CNN, while CAM requires a particular architecture.
 - ☐ (C) Because saliency maps show which parts of the image the model actually uses, a model whose saliency lies largely outside the anatomical region of interest is necessarily performing poorly on the task.
 - ☐ (D) TCAV explains a prediction in terms of human-defined concepts by measuring the directional derivative of the class score along a concept activation vector, and can be used to spot biases in a data set.
9. Which of the following statements about self-attention, as it was built up in the lecture, is **incorrect**?
- ☐ (A) Parameter-free dot-product attention, $Y = \text{Softmax}(XX^T)X$, has no capacity to learn from data, and every feature within a token plays an equal role in determining the attention coefficients.
 - ☐ (B) Introducing a single learnable matrix U and computing $Y = \text{Softmax}(XUU^TX^T)XU$ still yields a symmetric matrix of attention scores, which is why separate key and query projections are introduced.
 - ☐ (C) The scaling by $1/\sqrt{D_K}$ is used because the gradients of the softmax become very small for very large or very small inputs; if the entries of Q and K have unit variance, the entries of QK^T have variance D_K .
 - ☐ (D) In multi-head self-attention the outputs of the individual heads are averaged, so that the output dimension equals D_V irrespective of the number of heads.
10. In the Noise2Noise setting you have no clean images at all. For each input x_n you only have several *noisy* versions y_n^m of the underlying clean image y_n , and you train with $\arg \min_{\theta} \sum_n \sum_m \|y_n^m - f(x_n; \theta)\|^2$. Under which condition does this give the same solution as supervised training on clean targets, in the limit of infinite data?
- ☐ (A) When the noise is multiplicative rather than additive.
 - ☐ (B) When the average of the noisy targets is the clean image, $\mathbb{E}[\hat{y} \mid x] = y$, because the prediction that minimizes an ℓ_2 loss is the mean of the targets.
 - ☐ (C) When at least half of the training targets are clean images.
 - ☐ (D) When an adversarial loss is added to the ℓ_2 loss, so that the outputs are forced to look realistic.
 - ☐ (E) Never — training on noisy targets always biases the solution towards the noise.
11. Consider a classification network that takes grey-scale input images of size 64×64 with one input channel. All convolutions are *valid* (no padding, $P = 0$), and the network uses strided convolutions instead of pooling:
- **Conv1**: 16 kernels with kernel size $K = 5 \times 5$ and stride $S = 2$ in each direction.

- **Conv2**: 32 kernels with $K = 5 \times 5$ and $S = 2$.
- **Conv3**: 64 kernels with $K = 3 \times 3$ and $S = 2$.

Recall the channel size of a strided, valid convolution: $N_l = \lceil (N_{l-1} - k_1)/s_1 \rceil$. Circle True or False:

- (a) True False The output of Conv3 has a spatial size of 5×5 .
 - (b) True False The global receptive field of a neuron in Conv3 is 13×13 .
 - (c) True False The global stride of a neuron in Conv3 is 6×6 , since the three strides of 2 add up.
 - (d) True False The total number of convolutional kernel weights (excluding the biases) is $5 \cdot 5 \cdot 1 \cdot 16 + 5 \cdot 5 \cdot 16 \cdot 32 + 3 \cdot 3 \cdot 32 \cdot 64 = 31\,632$.
12. Which of the following statements on restoration, synthesis and self-supervised learning are correct (multiple answers are possible)?
- ☐ (A) Losses of the MSE type tend to produce over-smoothed restorations, which is one motivation for perceptual losses, where the two images are passed through a separately trained network and their intermediate features are compared.
 - ☐ (B) In CycleGAN-style synthesis from unpaired data, the cycle-consistency term requires that mapping a source image to the target domain and back returns the original image.
 - ☐ (C) In contrastive learning, the two views of an image have to be produced by geometric transformations only, since intensity changes would destroy the surrogate label.
 - ☐ (D) A masked autoencoder uses a larger encoder on the visible parts of the image and a smaller decoder to reconstruct the masked parts; the trained encoder is then used as a backbone for downstream tasks.
 - ☐ (E) Adversarial losses compare individual pairs of images, which is what makes them suitable for unpaired data sets.